

Psychometric validation of EQ-5D-Y in children with asthma and their caregivers

Wenjing Zhou¹, Bo Ding¹, Zhihao Yang², Michael Herdman³, Bin Wu¹, Pei Wang⁴, Jan Busschbach⁵, Nan Luo⁶, and Yanming Lu¹

¹Shanghai Jiao Tong University School of Medicine Affiliated Renji Hospital

²Guizhou Medical University

³Office of Health Economics

⁴Fudan University School of Public Health

⁵Erasmus MC Afdeling Psychiatrie

⁶National University Singapore Saw Swee Hock School of Public Health

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Abstract

Background The child-friendly EQ-5D (EQ-5D-Y) has been available in both a 3-level (Y-3L) and a 5-level (Y-5L) version. The study aimed to assess the measurement properties of both versions in paediatric patients with asthma and their caregivers. **Methods** Patients and their caregivers completed the self-complete and proxy Y-5L and Y-3L questionnaires, respectively, twice in two consecutive visits. Test-retest reliability and patient-caregiver agreement of Y-3L index, Y-5L level sum score (LSS) and EQ Visual Analogue Scale (VAS) were assessed using intraclass correlation coefficient (ICC). Known-groups validity compared scores of patients with different asthma control statuses. Responsiveness to improved overall health and asthma control was assessed using standardized effect size (SES). **Results** Seventy-six patients (age range: 8-16.7 years) and their caregivers participated. The ICC for test-retest reliability of the Y-3L, Y-5L and EQ VAS scores ranged from 0.744 to 0.898 for the self-complete version and 0.525 to 0.767 for the proxy version. As hypothesized, Y-3L, Y-5L and EQ VAS scores derived from both EQ-5D-Y versions indicated that well or partially controlled patients had better health than uncontrolled patients. SES values ranged from 0.20 to 1.40 for self-complete version and 0.39 to 1.26 for proxy version. ICC for patient-caregiver agreement ranged from 0.556 for Y-5L LSS to 0.833 for Y-3L index. **Conclusion** The results support the reliability, validity, and responsiveness of both the self-complete and proxy versions of the two EQ-5D-Y questionnaires for assessing health outcomes of asthmatic children. Moreover, the self-complete and proxy-reported versions appear to have good agreement.

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Informative title: Testing construct validity, reliability, and responsiveness of EQ-5D-Y in children with asthma and their caregivers

Running title: Validation of EQ-5D-Y in asthmatic children

Wenjing Zhou^{1,2*}, Bo Ding¹, Zhihao Yang³, Michael Herdman^{6,7}, Bin Wu⁴, Pei Wang⁵, Jan Busschbach², Nan Luo⁷, Yanming Lu^{1*}

¹ Department of Paediatrics, Renji Hospital, School of Medicine, Shanghai Jiaotong University, China;

² Department of Psychiatry, Section Medical Psychology and Psychotherapy, Erasmus Medical Center, Rotterdam, The Netherlands;

³ Health Services Management Department, Guizhou Medical University, China;

⁴ Medical Decision and Economic Group, Department of Pharmacy, Ren Ji Hospital, School of Medicine, Shanghai Jiaotong University, China;

⁵ School of Public Health, Fudan University, Shanghai, China;

⁶ Office of Health Economics, London, UK;

⁷ Saw Swee Hock School of Public Health, National University of Singapore, Singapore

*Correspondence:

Wenjing Zhou

Department of Paediatrics, Ren Ji Hospital, School of Medicine, Shanghai Jiaotong University, PR China

No. 2000, Jingyue Road, Shanghai, China

Email: zhoujing2188@163.com

Telephone: +31685281032

Yanming Lu

Department of Paediatrics, Ren Ji Hospital, School of Medicine, Shanghai Jiaotong University, PR China

No. 2000, Jingyue Road, Shanghai, China

Email: luyanming@renji.com

Telephone: +8613611704368

Abstract

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Methods Patients and their caregivers completed the self-complete and proxy Y-5L and Y-3L questionnaires, respectively, twice in two consecutive visits. Test-retest reliability and patient-caregiver agreement of Y-3L index, Y-5L level sum score (LSS) and EQ Visual Analogue Scale (VAS) were assessed using intraclass correlation coefficient (ICC). Known-groups validity compared scores of patients with different asthma control statuses. Responsiveness to improved overall health and asthma control was assessed using standardized effect size (SES).

Results Seventy-six patients (age range: 8-16.7 years) and their caregivers participated. The ICC for test-retest reliability of the Y-3L, Y-5L and EQ VAS scores ranged from 0.744 to 0.898 for the self-complete version and 0.525 to 0.767 for the proxy version. As hypothesized, Y-3L, Y-5L and EQ VAS scores derived from both EQ-5D-Y versions indicated that well or partially controlled patients had better health than uncontrolled patients. SES values ranged from 0.20 to 1.40 for self-complete version and 0.39 to 1.26 for proxy version. ICC for patient-caregiver agreement ranged from 0.556 for Y-5L LSS to 0.833 for Y-3L index.

Conclusion The results support the reliability, validity, and responsiveness of both the self-complete and proxy versions of the two EQ-5D-Y questionnaires for assessing health outcomes of asthmatic children. Moreover, the self-complete and proxy-reported versions appears to have good agreement.

Keywords: asthma, construct validity, EQ-5D-Y, reliability, responsiveness

Introduction

Clinical measures in children with asthma focus on symptoms and pulmonary function which do not necessarily reflect how the condition impacts on children’s lives. There is a trend towards more formally use of children’s self-perceived health status, both physically and psychologically, when monitoring patient outcomes, assessing the impacts of treatment, or, potentially, when deciding on treatment options in paediatric asthma.¹ Moreover, international disease management guidelines recommend considering patient-centered goals and preferences when developing a treatment plans for asthma.² Patient-centered goals and preferences in fact are often taken into account in clinical practice when physicians discuss treatment options with patients and with parents of very young children. However, the discussion between physician and patient regarding patient-centered goals and preferences is usually not facilitated with formal assessment of Patient-Reported Outcome Measures (PROMs). RPOMs are standardized disease-specific or generic questionnaires that are developed for assessing quality of clinical care, treatment effects, and patient outcomes from patients’ own perspectives. There is an increase in routine collection of PROM data and use of such data for managing patients with chronic diseases including paediatric asthma.³

The EQ-5D-Y is a generic PROM assessing the health-related quality of life (HRQoL) of children and adolescents.⁴ It is developed from the EQ-5D-3L, a widely used generic HRQoL instrument for adults.⁵⁻⁸ As a ‘youth’ version, the EQ-5D-Y retains the five-dimension, three-level structure of the EQ-5D-3L,⁹ but uses child-friendly vocabulary. Notably, the EQ-5D-Y comes in two forms: a self-complete version that can be completed by children aged 8 years or above, and a proxy version that can be completed by parents or caregivers if a child is not able to complete the version by themselves for reasons such as very young age and suffering from learning problems, behavioral disorders or neurocognitive conditions.¹⁰ The EQ-5D-Y has demonstrated good measurement properties in many paediatric conditions, such as kidney disease, congenital physical disability, idiopathic arthritis and type 1 diabetes mellitus.¹¹⁻¹⁴ Some research investigated into the reliability and known-groups validity of the self-complete Y-3L in asthmatic children in the Netherlands, Sweden, Spain and South Korea.¹⁵⁻¹⁹ Recently, a 5-level EQ-5D-Y (EQ-5D-Y-5L) was developed from the EQ-5D-Y by increasing the number of levels to 5 for each health dimension.²⁰ EQ-5D-Y-5L (hereafter referred to as Y-5L and EQ-5D-Y as Y-3L) is the counterpart of the adult EQ-5D-5L which was developed from EQ-5D-3L. The Y-5L has been validated in some paediatric conditions, such as idiopathic scoliosis,²¹ haematological diseases,²² and psychological disorder.²³ To the best of our knowledge, the measurement properties of Y-5L in children with asthma has not be evaluated, though a small group of asthma patients were included in a validation study of the instrument in South Africa.²⁴

This study aimed to concurrently assess the reliability, validity, and responsiveness of both the self-complete and proxy versions of the Y-3L and Y-5L in Chinese children and adolescents with asthma. We also assessed the inter-rater agreement between the self-complete and proxy versions.

Methods

Sampling

Paediatric outpatients with asthma and their caregivers who visited Renji Academic Hospital in Shanghai, China for routine consultation from November 2018 to July 2019 were invited to participate in this study. The inclusion criteria for patients were: 1) a diagnosis of asthma by a specialist; 2) aged between 8-17 years; and 3) ability to converse in Chinese. Patients who were not well enough to complete questionnaire-based surveys, or whose legal guardians did not give written informed consent were excluded. The inclusion criteria for caregivers were: 1) adult family member of an eligible patient; 2) being with the patient in the outpatient clinics on the day of the survey; 3) ability to converse in Chinese; 4) being cognitively able to complete questionnaires. The study was approved by the institutional medical ethical review board of Shanghai Jiaotong University (Project IRB number: 2018087).

Procedures

All consenting patients and caregivers were asked to complete an electronic survey form independently in the paediatric clinics using a tablet PC. The patient survey form included: 1) the Y-5L including the EQ VAS; 2) a 5-point Overall Health Assessment (OHA) question, and 3) the Y-3L without the EQ VAS. Caregivers were asked to complete: 1) questions on their patients’ socio-demographic and clinical characteristics, including birth date, gender, disease duration, and educational level; 2) the Y-5L questionnaire (proxy version); 3) the OHA question (proxy version) for their child; 4) questions on caregiver demographics, including relationship with the patient, age, educational attainment, residential area, marriage status, and monthly household income, and 5) the Y-3L proxy version without the EQ VAS. All participants were invited to fill in the same questionnaire again in one of their follow-up visits to the outpatient clinics. The follow-up survey forms were the same as the baseline survey forms except that the order of the Y-5L and Y-3L questionnaires was swapped and the demographic questions were omitted. On the day of the survey, the level of asthma control was assessed by a clinician according to the GINA asthma control criteria.²

Instruments

The Y-3L questionnaire assesses respondents’ HRQoL on the day of the survey using five questions each targeting a different dimension. The dimensions are mobility (walking about), looking after myself (washing or dressing myself), doing usual activities (for example, going to school, hobbies, sports, playing, doing things with family or friends), having pain or discomfort, and feeling worried, sad or unhappy. Each dimension has three response options corresponding to the severity levels of no problems, some problems, and a lot of problems. In comparison, Y-5L has five response options corresponding to the levels of no problems, a little bit of a problem, some problems, a lot of problems, and extreme problems/cannot.²⁰ Both questionnaires also include a vertical, 20-cm-long, hash-marked visual analogue scale (EQ VAS) anchored by 0 (the worst imaginable health) at the bottom and 100 (the best imaginable health) on the top on which respondents rate their overall health on the day of the survey.²⁵ There are two proxy versions of EQ-5D-Y. The proxy version 1, which was used in this study, asks caregivers to provide their impression of the patient’s health on the day of survey. The Chinese (for China) Y-3L and Y-5L questionnaires used in this study were developed through a combination of iterative translation and cultural adaptation procedures as recommended by the EuroQol Group.²⁶ The Y-3L was the approved version of the instrument, whereas the Y-5L was still considered an experimental version.

The Overall Health Assessment question (OHA) has been shown to be a valid measure of subjective health in children and adolescents.²⁷ The question in the present study was phrased as ‘How is your overall health today? Is it excellent, good, fair, poor or very poor?’ The proxy version was phrased so as to collect caregiver’s impression of the patient’s overall health on the day of survey.

Global Initiative for Asthma (GINA) criteria for asthma control are based on cough and wheezing symptoms during the past four weeks. The criteria include: 1) symptoms happening in daytimes > 2 times a week; 2) night awakening due to asthma; 3) emergency relief medication used > 2 times a week; 4) activity limitation due to asthma. A good control is defined as satisfying none of the above criteria, a partial control satisfying 1-2 criteria, and a poor control satisfying 3-4 criteria.²

Data analysis

We assessed the test-retest reliability, known-groups validity, and responsiveness of the Y-3L preference-weighted index score, Y-5L level sum score (LSS), and EQ VAS score (collectively referred to as ‘EQ-5D-Y scores’) generated from the self-complete and proxy versions of the Y-3L and Y-5L questionnaires separately.

We used the Chinese Y-3L value set²⁸ to calculate the preference-weighted index scores for Y-3L. This score ranges from -0.088 to 1, with higher values indicating better health utility. A LSS was calculated as a proxy of the preference-weighted index for Y-5L since no value set was available. We coded “no problem” as 1, “a little bit problems” as 2, “some problems” as 3, “a lot of problems” as 4, and “extreme problems” as 5 for all five EQ-5D-Y dimensions. Thus, the LSS for a reported health state 11223 is $1+1+2+2+3 = 9$.

Test-retest reliability of the EQ-5D-Y scores was assessed using data from patients whose overall health status at the baseline and follow-up surveys remained the same using the Overall Health Assessment (OHA) question. Intraclass correlation coefficient (ICC) was used, with an ICC [?] 0.7 being considered as acceptable test-retest reliability.²⁹ Patient-caregiver agreement also was assessed for both Y-3L and Y-5L using baseline data from all patient-caregiver dyads by ICC.

Construct validity was assessed using data collected at baseline by determining whether the EQ-5D-Y scores differed between patients known to differ in their asthma control, i.e. patients being well controlled/partially controlled and uncontrolled according to GINA. We hypothesized that well controlled/partially controlled patients would have better scores (i.e. higher index and EQ VAS scores and lower LSS) than uncontrolled patients. Both two-sample t test and non-parametric Mann-Whitney U test were used to determine whether there were statistically significant differences between the groups and Cohen's d effect size (ES) was also used to assess the relative efficiency of the EQ-5D-Y scores in discriminating between patients in 'good' and 'poor' disease control.

The responsiveness of the EQ-5D-Y scores was assessed using data collected from patients whose health status improved from baseline to follow-up visits, according to their OHA or GINA classification. Standardized effect size (SES) (difference of mean/baseline SD) was calculated, with an SES of 0.20 to 0.49, 0.5 to 0.79 and 0.8+ being considered as representing modest, moderate and good responsiveness, respectively.³⁰

All analyses were conducted using the statistical package SPSS (IBM SPSS Statistics, Version 26.0, IBM Corp).

Results

A total of 76 paediatric outpatients and their caregivers participated in the baseline survey (see Table 1). Seventy (92.1%) patients and their caregivers also completed the follow-up surveys. The mean (SD) time between the two assessments was 42.0 (21.3) days (range: 5-110 days). According to clinician's GINA asthma control assessment, the disease status of 26 (37.1%) patients was improved, 39 (55.7%) remained the same and 5 (7.1%) was aggravated at the follow-up survey. According to patients and proxies, the overall health of 24 (34.3%) and 26 (37.1%) patients, respectively, improved from baseline to follow-up, while 39 (55.7%) and 38 (54.3%) remained unchanged, and 7 (10.0%) and 6 (8.6%) patients deteriorated.

The baseline health status of the patients as described by the two EQ-5D-Y questionnaires is shown in Table 2. In each dimension, over two-thirds of the patients reported no problems and approximately half of the patient sample reported no problems in all five dimensions. For both classification systems, the 'looking after myself' dimension showed the same highest proportion of no problems (98.7%), whereas the 'pain/discomfort' dimension exhibited the lowest proportion of no pain or discomfort (Y-3L: 63.2%; Y-5L: 64.5%). The self-complete Y-5L showed a slightly higher ceiling effect than the Y-3L. On average, proxies described the patients's health similarly to the patients themselves on both the Y-3L and the Y-5L descriptive systems. The mean self-rated EQ VAS score was slightly higher than the proxy ratings (85.5 versus 83.5, $p = 0.069$), while the mean Y-3L index and Y-5L LSS scores based on patients and their proxies were very similar.

To examine test-retest reliability, we selected the 39 patients and 38 caregivers who rated the patients' overall health as unchanged at follow-up. The agreements in the Y-5L LSS, EQ VAS and Y-3L index score between the two assessments by patients was good (ICC=0.898, 0.802 and 0.744, respectively). The baseline-follow-up agreement for the proxy versions was slightly lower (ICC=0.767, 0.661, and 0.525, respectively), suggesting that the self-complete version was more reliable compared to the proxy version (Table 3).

The known-groups validity results are displayed in table 4. In general, patients who were well or partially controlled exhibited higher Y-3L index and VAS scores, as well as lower LSS (indicating better health) compared to poorly controlled patients, though the difference in some of the scores was not statistically

significant. The difference of means in the Y-3L index score and the Y-5L LSS was similar, with ES ranging from 0.58 to 0.60 for the self-complete version ($p[?]0.024$), and from 0.41 to 0.45 for the proxy version ($p[?]0.057$), respectively. Regarding the EQ VAS, the ES values were much higher rated by parents ($ES=0.89$) than by patients ($ES=0.34$, $p=0.144$).

Table 5 presents the mean baseline and follow-up Y-3L index score, Y-5L LSS and EQ VAS for patients who experienced improvements in their overall health and asthma control over time. Generally, all the EQ-5D-Y scores showed improved, although some of the changes were not statistically significant. For instance, the mean (SD) baseline and follow-up Y-3L index scores of patients whose GINA ratings improved ($n=25$) were 0.93 (0.07) and 0.95 (0.15) ($p = 0.716$), respectively. The SES values for the changes in Y-5L LSS (range: 0.19-0.78) were mostly lower than those in Y-3L index score (range: 0.28-1.00). While the SES values for the score changes for the proxy versions (range: 0.39-0.88) were slightly higher than the self-complete (range: 0.19-0.28) based on GINA categories.

In terms of the patient-caregiver agreement (Table 3), the Y-3L based on EQ index score demonstrated better reliability ($ICC=0.833$) compared to the Y-5L based on LSS ($ICC=0.556$), and good for EQ VAS as well ($ICC=0.810$).

Discussion

The self-complete EQ-5D-Y-3L has demonstrated good measurement properties in children with asthma.¹⁵⁻¹⁹ However, little evidence is available about the performance of its proxy version or the EQ-5D-Y-5L in this patient population. A recent study comparing the construct validity and test-retest reliability of the Y-3L and Y-5L included a minor sample of asthma patients ($n<20$), but lacked an independent analysis for the Y-5L.²⁴ To the best of our knowledge, our study is the only one that has examined the measurement properties of both the self-complete and proxy versions of EQ-5D-Y-5L in this particular patient population. The main finding of our study was that, both instruments (Y-3L and Y-5L) and versions (self-complete and proxy) demonstrated reasonable test-retest reliability, construct validity, and responsiveness, and that the self-complete and proxy versions exhibited good agreement. Notably, our study is the very first study of the responsiveness of EQ-5D-Y-5L in asthmatic children. The responsiveness analysis in the subgroups with a disease improvement showed the values increased. These good measurement properties of the self-complete version of Y-3L observed in our study are generally consistent with similar studies in the literature. For example, our findings on test-retest reliability and known-groups validity were similar to those from a recent Spanish study ($ICC: 0.70$ to 0.81 ; $ES: 0.38$ to 0.66) of persistent asthmatic children.¹⁹

Our study also provided support for the reliability, construct validity and responsiveness (to improvement) of the proxy version of both Y-3L and Y-5L. It is worth noting that the proxy version demonstrated slightly lower reliability compared to the corresponding self-complete version. However, it showed a tendency of better responsiveness than the self-complete version, especially in terms of the Y-5L LSS and the EQ VAS. This difference could be attributed to caregivers, who are adults, being more able to understand and respond to more complex rating scales compared to children and adolescents. The EQ VAS requires a higher cognitive ability to complete and children are not fully up to use it.³¹ It could also be due to patients' coping or adaptation. Even though patients experience some health problems, they might have developed strategies to mitigate the influence and feel they are fairly healthy. The caregivers, however, did not develop any coping strategies and therefore are more sensitive to change in patients' health. This, together with the fact that there are significant ceiling effects with the EQ-5D-Y descriptive system but not the EQ VAS, suggests that EQ VAS is a useful proxy assessment measure particularly for children with mild impairment in health-related quality of life.³²

In this study, both the self-complete and proxy EQ-5D-Y versions did not show an obvious reduction in ceiling effect from Y-3L to Y-5L. Although there is a lack of evidence from previous studies in children with asthma, our findings align with other comparisons between the 3-level and 5-level versions of the EQ-5D-Y

in children in China: the Y-5L had higher ceiling effects than Y-3L for short sighted and overweight pupils.³³ Although the Y-5L was developed with the aim of increasing the instrument's sensitivity and reducing the ceiling effect of the Y-3L, evidence on whether it does provide such an increase in sensitivity is mixed.^{22,34} Nonetheless, the Y-5L shows at least comparable psychometric performance to the Y-3L, and there is some evidence that Y-5L is preferred by children themselves.²⁰

The good measurement properties of EQ-5D-Y revealed in our study suggest that it can be a useful HRQoL instrument or PROM for paediatric asthmatic patients. In addition to be a PROM, EQ-5D-Y is specifically designed to inform policy makers about the burden of disease and cost effectiveness analysis from a societal perspective. In the current practice of economic evaluation of new health technologies, evaluation of effectiveness involves measurement and valuation of HRQoL using a preference-based measure. EQ-5D-Y is one of the very few such measures for paediatric economic evaluation. Therefore, a valid EQ-5D-Y for asthma will be an useful instrument for health policy making for this prevalent illness in children and adolescents.

This study had several limitations. First, the sample size was only moderate which prevented an analysis for patients with worsened health status. Second, all participants were recruited from one hospital in Shanghai, which has a high socio-economic status. As asthma and the control of asthma is related to social economic status,² it might be that results presented in this study might not be generalisable to other regions. Third, the Y-5L was assessed using a level sum score which may behave differently from a preference-based index score. Therefore, the findings for Y-5L in this study should be treated as preliminary and the further studies are needed when value sets of Y-5L are available in the future.

In conclusion, this study contributes to the growing body of evidence about the measurement properties of the youth version of the EQ-5D. The results show that both the self-complete and proxy versions of the EQ-5D-Y-5L and EQ-5D-Y-3L appear to be reliable, valid, and responsive for assessing health outcomes of asthmatic children. Although some differences were found, neither of the two questionnaires showed clear superior performance over the other in this patient group.

Author contributions

Nan Luo, Yanming Lu, Zhihao Yang, Pei Wang, Bin Wu, Michael Herdman and Jan Busschbach conceived of the study, participated in the design, and made substantial contribution to the intellectual content of the manuscript. Wenjing Zhou, Bo Ding and Nan Luo participated in the acquisition and interpretation of data, performed statistical analyses and writing of the manuscript. The first draft of the manuscript was written by Wenjing Zhou and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript and agreed to be accountable for all aspects of the work.

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Conflict of interest

All authors declare that they have no conflicts of interest.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request

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